

SAMPLING AWARE ANCESTRAL STATE INFERENCE (SAASI)

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PhylogeogRaphy Workshop

Day 4

1 BACKGROUND

2 SAMPLING AWARE ANCESTRAL STATE INFERENCE

3 PRACTICAL CONSIDERATIONS

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BACKGROUND

With the exception of the multi type birth death model (MTBD), all of the methods we have considered have assumed that we are equally likely to observe a lineage regardless of its state

This is not typically reasonable, and it can be shown to produce biased estimates of ancestral states¹

¹Pengyu Liu et al. "The impact of sampling bias on viral phylogeographic reconstruction". In: *PLOS Global Public Health* 2.9 (2022), e0000577.

BACKGROUND

Sampling aware ancestral state inference² incorporates state-dependent sampling rates into a BiSSE-style likelihood calculation

²Yexuan Song et al. "Sampling Aware Ancestral State Inference". In: *bioRxiv* (2025), pp. 2025–05.

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BIRTH-DEATH SAMPLING MODEL

The tree is assumed to arise through a birth-death sampling model

Processes include

- diversification rates λ_{ijk} (lineage in state i can split into two lineages in completely different states) “cladogenic”
- death at rate μ_i
- migration at rate q_{ij}
- lineage are sampled at rate ψ_i

This gives rise to a likelihood of the type we have seen repeatedly this week (backward Kolmogorov equations integrated up branches)

However, probabilities for particular edges are tracked (rather than subtrees descending from edges)

BIRTH-DEATH SAMPLING MODEL

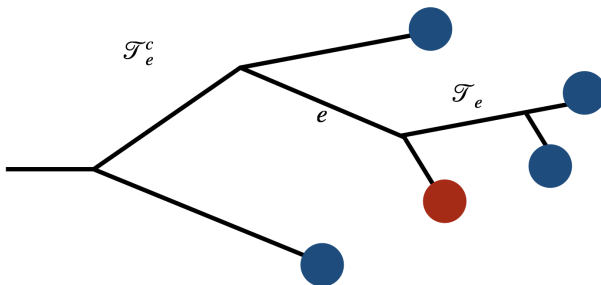
The variable τ measures time going into the past from the tips, and t measures time going into the future from the root

A key variable is $A_{e,i}(\tau) = P(Y_e = i | \mathcal{T}, \theta)$, the probability that edge e is in state i at time τ , given the tree $\mathcal{T}$ and the parameters of the birth-death sampling model, θ

This is slightly different from the $D_{Ni}(t)$'s we saw in BiSSE and birth-death models, which is the probability of observing the subtree descending from node N and that edge N in resides in state i at time τ

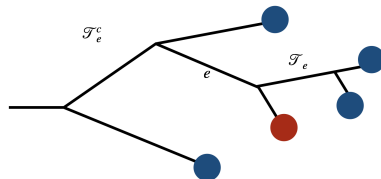
BIRTH-DEATH SAMPLING MODEL

The probability an edge resides in a particular state depends on the tree descending from that edge, as well as the tree that descends from above (and in a state-dependent manner):



$$A_{e,i}(\tau) = P(Y_e(\tau) = i | \mathcal{T}, \theta),$$
$$\frac{P(Y_e(\tau) = i | \mathcal{T}_e^c(\tau), \theta) P(\mathcal{T}_e(\tau) | Y_e(\tau) = i, \theta)}{\sum_j P(Y_e(\tau) = j | \mathcal{T}_e^c(\tau), \theta) P(\mathcal{T}_e(\tau) | Y_e(\tau) = j, \theta)}$$

BIRTH-DEATH SAMPLING MODEL

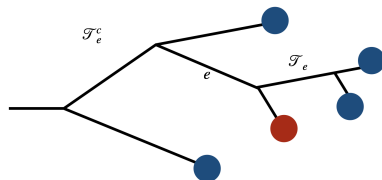


$$A_{e,i}(\tau) = P(Y_e(\tau) = i | \mathcal{T}, \theta),$$

$$\frac{\tilde{D}_{ei}(t) D_{ei}(\tau)}{\sum_j \tilde{D}_{ej}(t) D_{ej}(\tau)}$$

$D_{ei}(\tau) = P(\mathcal{T}_e(\tau) | Y_e(\tau) = i, \theta)$ is as in BiSSE and MTBD, but
 $\tilde{D}_{ei}(\tau) = P(Y_e(T - \tau) = i | \mathcal{T}_e^c, \theta)$ corresponds to $\mathcal{T}_e^c$

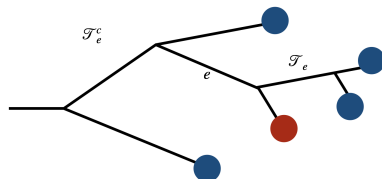
BIRTH-DEATH SAMPLING MODEL



$$A_{e,i}(\tau) = P(Y_e(\tau) = i | \mathcal{T}, \theta),$$
$$\frac{\tilde{D}_{ei}(t) D_{ei}(\tau)}{\sum_j \tilde{D}_{ej}(t) D_{ej}(\tau)}$$

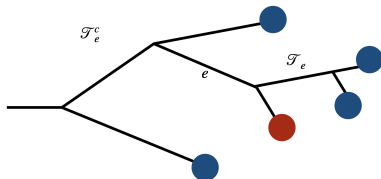
Calculation of this quantity requires integrating from tips to root (postorder traversal/pruning), and then from root to tips (preorder traversal)

BIRTH-DEATH SAMPLING MODEL



$$A_{e,i}(\tau) = P(Y_e(\tau) = i | \mathcal{T}, \theta),$$
$$\frac{\tilde{D}_{ei}(t) D_{ei}(\tau)}{\sum_j \tilde{D}_{ej}(t) D_{ej}(\tau)}$$

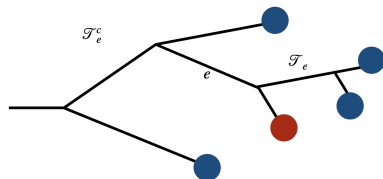
Note that this immediately gives the the probability edge e resides in a particular state (and this can also be calculated at internal nodes)

BACKWARD KOLMOGOROV EQUATIONS FOR $D_{ei}(\tau)$ 

At the start of branch e , we initialize $D_{ei}(\tau)$:

$$D_{ei} = \begin{cases} \psi_i & \text{if sampling,} \\ \sum_{ij} \lambda_{ijk} D_{e_1j} D_{e_2k} & \text{if birth} \end{cases}$$

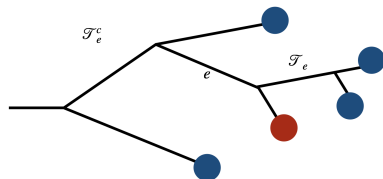
BACKWARD KOLMOGOROV EQUATIONS FOR $D_{ei}(\tau)$



We integrate $D_{ei}(\tau)$ up branch e according to

$$\begin{aligned} \frac{dD_{ei}}{d\tau} = & - \left(\sum_{jk} \lambda_{ijk} + \mu_i + \psi_i + \sum_{j \neq i} q_{ij} \right) D_{ei} \\ & + \sum_{jk} 2\lambda_{ijk} D_{ej} E_k + \sum_{j \neq i} q_{ij} D_{ej} \end{aligned}$$

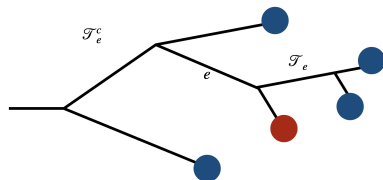
BACKWARD KOLMOGOROV EQUATIONS FOR $E_i(\tau)$



We integrate $E_i(\tau)$ up branch e according to

$$\begin{aligned} \frac{dE_i}{d\tau} = & \mu_i - \left(\sum_{jk} \lambda_{ijk} + \mu_i + \psi_i + \sum_{j \neq i} q_{ij} \right) E_i \\ & + \sum_{jk} \lambda_{ijk} E_j E_k + \sum_{j \neq i} q_{ij} E_j \end{aligned}$$

ROOT CONDITION

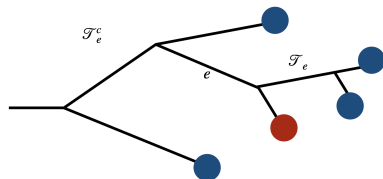


At the root,

$$A_{Ri}(T) = \frac{D_{Ri}(T)\pi_i}{\sum_j D_{Rj}(T)\pi_j},$$

and define $\tilde{D}_{Ri}(0) = \pi_i$, with the argument of $\tilde{D}_{Ri}$ measured in forward time from the root, and π_i is a “prior” probability

PREORDER TRAVERSAL

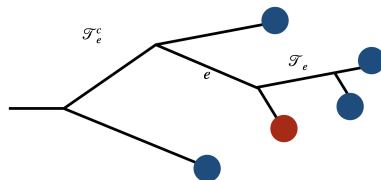


Then, we start at the root and integrate $\tilde{D}_{ei}$ down the branch toward the tips, in a preorder traversal (Forward Kolmogorov equations?)

At the end of a branch when lineages bifurcate, the $\tilde{D}_{ei}(t)$ on descendant edges are initialized according to

$$\tilde{D}_{ei}(t) = \frac{\sum_{j,k} 2\lambda_{j,i,k} \tilde{D}_{ej} D_{e'k}}{\sum_{j,l,k} \lambda_{j,l,k} \tilde{D}_{ej} D_{e'k}}$$

ANCESTRAL STATE ESTIMATE



The quantity

$$A_{e,i}(\tau) = P(Y_e(\tau) = i | \mathcal{T}, \theta),$$
$$\frac{\tilde{D}_{ei}(t) D_{ei}(\tau)}{\sum_j \tilde{D}_{ej}(t) D_{ej}(\tau)}$$

can be calculated at internal nodes or at points along edges; either way, it gives the probability that a particular location in the tree resides in state i , integrating over all other possibilities elsewhere in the tree; i.e. a marginal estimate.

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WHICH PARAMETERS CAN BE ESTIMATED?

Currently does not jointly estimate state-dependent sampling rates ψ_i alongside other birth-death model parameters

Currently assumes phylogeny $\mathcal{T}$ is fixed

Can be constrained to omit cladogenesis (lineage producing daughter lineages in different states), migration, or state-dependent differences in birth rates

Open question: currently, SAASI is not set up to jointly estimate all parameters. Future work could explore identifiability of sampling rates and other parameters